



Hunting Inside N-Quantiles of Outliers (Hino)

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Motivation

What is an outlier?

- ▶ This is a point that is inconsistent with the rest of the dataset.

Inconsistence have two effects on ML, in:

- ▶ Learning phase: Weaken the predictive power.
- ▶ Validation phase: Weaken the cross-validation score.

➡ They must be removed from these corpus.

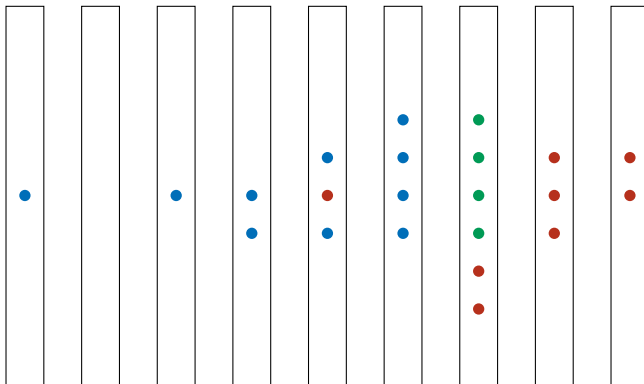
Research questions:

1. How do you characterize an outlier?
2. It's possible to have an automatic method?

➡ Validation on real and synthetic datasets.



How we define outliers



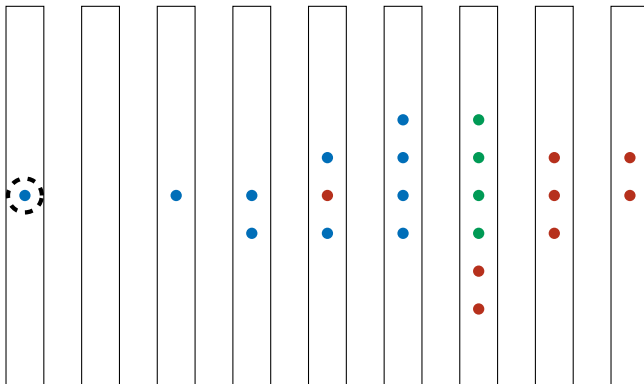
Quantiles must be:

- ▶ small enough to have close points;
- ▶ large enough to contain one per class.

Number of quantile is $\frac{n_p}{n_c+1}$.



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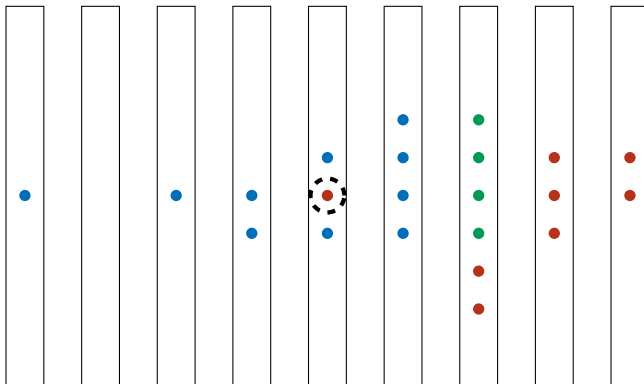
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Isolation of a point:

- ▶ In space.



How we define outliers



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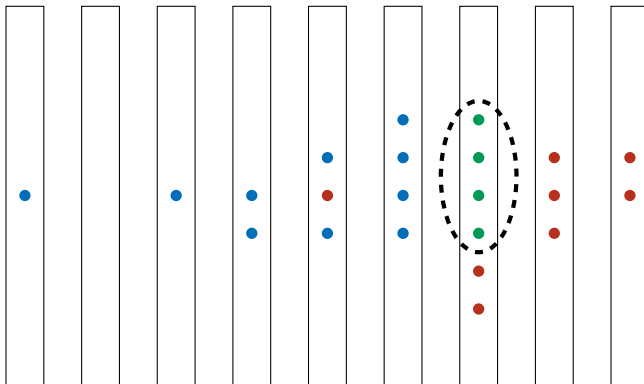
Number of quantile is $\frac{n_p}{n_c+1}$.

Isolation of a point:

- ▶ In space.
- ▶ From its peers.



How we define outliers



Quantiles must be:

- ▶ small enough to have close points;
- ▶ large enough to contain one per class.

Number of quantile is $\frac{n_p}{n_c+1}$.

Isolation of a point:

- ▶ In space.
- ▶ From its peers.

Can't be isolated if all points of a class are in one quantile.



Hino's algorithm (based on IQR)

Isolation tolerance:

- ▶ Being isolated 1 time $\nRightarrow$ being an outlier;
- ▶ A point can be isolated in several attributes.

➡ Set a limit: $\lfloor (0.0205 * \log_2(-1.623730 + n_c) + 0.062579) * n_a \rfloor$

How it works:

- ▶ Splits points into several quantiles for each attribute;
- ▶ **Isolation score of a point:** number of times where its class isn't present in the 2 adjacent quantiles;
- ▶ **Outlier:** isolation score $\geq$ tolerance limit;
- ▶ **Restarts with tolerance limit +1** if:
 - ▶ % outlier detected $>$ the % allowed.
 - ▶ a class is removed.



Experimentation objectives

Determine if Hino is:

- ▶ A **viable** method:
 - ▶ Removes a maximum of outliers.
 - ▶ Don't removes too much healthy points.
 - ▶ Doesn't eliminate a class.
- ▶ **Better** than the state-of-the-art methods:
IQR [11], SVM [2], Isolation Forest [7] and LOF [4].

The evaluation was realized on:

- | | |
|----------------------------------|--|
| ▶ 278 synthetic datasets: | ▶ 16 real datasets: |
| ▶ Large amount of sample. | ▶ Representative i.r. characteristics. |
| ▶ Outliers are known. | ▶ Outliers are unknown. |
| ▶ Bias i.r. of real world. | ▶ Real usage. |



Synthetic datasets

How they created:

- ▶ Generate with the algorithm used to create Madelon (from UCI) [6].
- ▶ Modify 5% of points and make them outliers:
 - ▶ Preserve class representativity.
 - ▶ Choose points with unanimity predicted by ML.
 - ▶ Change randomly their classes.

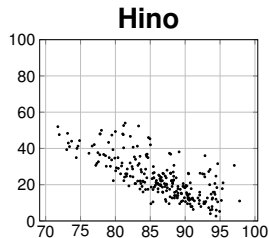
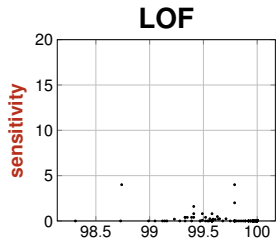
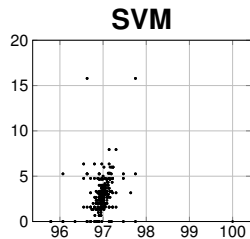
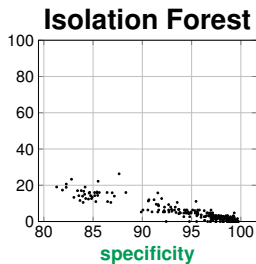
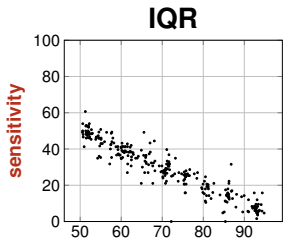
Two kinds of measure:

- ▶ **Sensitivity** → % correctly predicted outliers
- ▶ **Specificity** → % correctly predicted healthy points.

Samples	Attributes	Classes
250	10 ... 50, 25	2 ... 5
500	10 ... 70, 25, 75	2 ... 5
1 000	10 ... 100, 25, 75	2 ... 7
2 500	10 ... 100, 25, 75	2 ... 8
5 000	10 ... 100, 25, 75, 125	2 ... 11



Results on synthetic datasets



Real datasets

UCI Machine Learning Repository:

- ▶ Classification.
- ▶ Continuous values.
- ▶ Cover several combination of characteristics.

Two kinds of measures (with 6 ML):

- ▶ Cross-validation:
 - ▶ Detected outliers removed.
 - ▶ 70% train + 30% test
 - ➡ % good prediction.
- ▶ False positivity:
 - ▶ Healthy as train & Outliers as test.
 - ➡ % good prediction.

x10

x10

Datasets	Points	Attributes	Classes
annthyroid	7 200	21	3
breastW	683	9	2
cardio	2 126	21	10
glass	214	9	6
ionosphere	351	33	2
isolet	7 797	617	26
letter recognition	20 000	16	26
mammo graphy	11 183	6	2

Datasets	Points	Attributes	Classes
multiple features	2 000	619	10
musk	6 598	166	2
parkinson	756	752	2
pendigits	10 992	16	10
satimage2	6 435	36	6
shuttle	58 000	9	7
wine	178	13	3
wine quality	1 599	11	6



Example results on real datasets

Generally, Hino:

- ▶ Removes less points.
- ▶ Gives a better cross-validation.
- ▶ Gives a better false-positivity.
- ▶ Tolerance limit calculated is relevant.

Over 16 real datasets:

- ▶ 13 gives **Hino** as better solution.
- ▶ 2 is even between **Hino** and **LOF**.
- ▶ 1 is even between **Hino**, **LOF** and **SVM**.

Datasets	IQR		Isolation Forest		SVM		LOF		Hino		Best of Hino	
	Cross validation	False positive	Cross validation	False positive	Cross validation	False positive	Cross validation	False positive	Cross validation	False positive	Cross validation	False positive
cardio	81.40% ±1.94 56.30% ✗	60.35% ±0.19	81.23% ±1.26 16.04% ✓	72.71% ±0.35	82.06% ±1.50 3.15% ✓	71.67% ±0.30	82.45% ±1.69 1.69% ✓	71.67% ±3.15	83.16% ±1.30 3 5.90% ✓	39.72% ±0.59	84.76% ±1.30 2 12.61% ✓	43.77% ±0.57
pendigits	98.58% ±0.20 4.70% ✓	85.18% ±0.17	98.87% ±0.23 51.91% ✗	50.45% ±0.17	98.52% ±0.16 2.98% ✓	94.99% ±0.16	98.67% ±0.20 1.56% ✓	76.34% ±0.45	98.63% ±0.20 2 16.70% ✓	97.27% ±0.04	98.58% ±0.12 9 0.01% ✓	0.00% ±0.00

Hino never removes a class and is always better than **IQR**.



Conclusion & perspective

Hino is:

- ▶ A **viable** approach.
- ▶ A **better** approach than the state-of-arts ones.
- ▶ **Takes more time than IQR**: $O(n_p(n_a n_c + n_a + 1))$ vs $O(n_p * 2n_a)$
where n_p as number of points, n_a as number of attributes and n_c as number of classes.

Can be **improved** by changing the way of computation meta-parameters:

- ▶ Better tolerance limit equation **and/or** refines maximal outliers allowed.
- ▶ Tests other equation for the number of quantile.

Use **Hino** as an incremental approach:

- ▶ Use a trained **Hino** model to detect outliers.
- ▶ Compare with other incremental methods.



Thanks for your attention!

Source code can be found here: <https://github.com/JessyColonval/Hino>

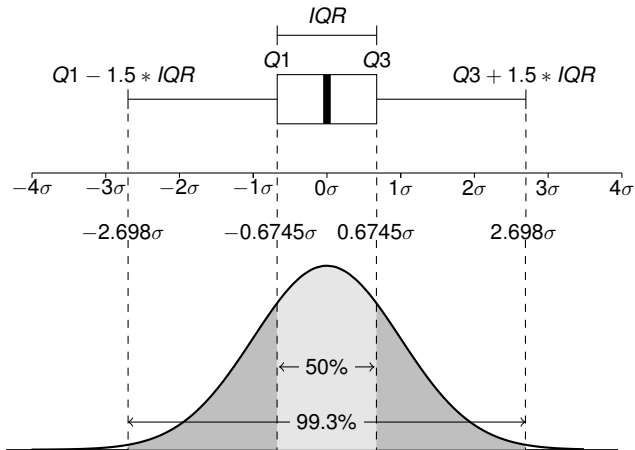


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Bonus: IQR algorithm



- ▶ Divides the dataset into 4 quantiles;
- ▶ No tolerance limit;
- ▶ Points with extreme values are outliers.

